Test Series No.

1.12

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 19/03/2026

ANSWER KEY

1.	(1)	46.	(4)	91.	(2)	136.	(3)
2.	(2)	47.	(1)	92.	(4)	137.	(1)
3.	(4)	48.	(3)	93.	(3)	138.	(3)
4.	(3)	49.	(4)	94.	(2)	139.	(2)
5.	(1)	50.	(3)	95.	(4)	140.	(3)
6.	(4)	51.	(3)	96.	(1)	141.	(3)
7.	(3)	52.	(2)	97.	(4)	142.	(2)
8.	(3)	53.	(1)	98.	(2)	143.	(2)
9.	(1)	54.	(3)	99.	(3)	144.	(1)
10.	(4)	55.	(2)	100.	(4)	145.	(1)
11.	(2)	56.	(2)	101.	(1)	146.	(2)
12.	(3)	57.	(2)	102.	(3)	147.	(3)
13.	(3)	58.	(2)	103.	(2)	148.	(1)
14.	(2)	59.	(1)	104.	(4)	149.	(2)
15.	(1)	60.	(1)	105.	(3)	150.	(1)
16.	(2)	61.	(2)	106.	(1)	151.	(2)
17.	(2)	62.	(1)	107.	(1)	152.	(3)
18.	(3)	63.	(2)	108.	(2)	153.	(4)
19.	(1)	64.	(4)	109.	(3)	154.	(3)
20.	(3)	65.	(3)	110.	(4)	155.	(3)
21.	(2)	66.	(4)	111.	(2)	156.	(2)
22.	(3)	67.	(2)	112.	(2)	157.	(1)
23.	(1)	68.	(2)	113.	(1)	158.	(4)
24.	(2)	69.	(1)	114.	(3)	159.	(2)
25.	(3)	70.	(2)	115.	(2)	160.	(1)
26.	(4)	71.	(1)	116.	(1)	161.	(3)
27.	(2)	72.	(1)	117.	(4)	162.	(3)
28.	(1)	73.	(3)	118.	(3)	163.	(4)
29.	(3)	74.	(2)	119.	(2)	164.	(2)
30.	(3)	75.	(2)	120.	(3)	165.	(2)
31.	(2)	76.	(3)	121.	(3)	166.	(2)
32.	(4)	77.	(2)	122.	(1)	167.	(3)
33.	(3)	78.	(3)	123.	(2)	168.	(4)
34.	(3)	79.	(1)	124.	(3)	169.	(3)
35.	(2)	80.	(2)	125.	(2)	170.	(3)
36.	(3)	81.	(2)	126.	(4)	171.	(4)
37.	(3)	82.	(1)	127.	(3)	172.	(1)
38.	(3)	83.	(4)	128.	(1)	173.	(2)
39.	(3)	84.	(4)	129.	(2)	174.	(3)
40.	(1)	85.	(4)	130.	(4)	175.	(1)
41.	(1)	86.	(3)	131.	(2)	176.	(1)
42.	(2)	87.	(2)	132.	(3)	177.	(2)
43.	(1)	88.	(4)	133.	(3)	178.	(4)
44.	(3)	89.	(1)	134.	(1)	179.	(1)
45.	(4)	90.	(2)	135.	(3)	180.	(2)

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Hints & Solutions

1. (1)

The mass of the hanging part is $m_1 = \left(\frac{M}{L}\right)x.$

The weight of this hanging part provides the pulling force: $F_{pull} = m_1 g = \left(\frac{M}{L}\right)xg.$

The length of the chain resting on the table (L-x). The mass of the part on the table is

$m_2 = \left(\frac{M}{L}\right)(L-x).$

The normal reaction is $N = m_2 g = \left(\frac{M}{L}\right)(L-x)g.$

The maximum static friction resisting the pull is

$f_{max} = \mu N = 0.25 = \left(\frac{M}{L}\right)(L-x)g.$

For the chain to just start slipping, $F_{pull} = f_{max}$:

$$= \left(\frac{M}{L}\right)xg = 0.25\left(\frac{M}{L}\right)(L-x)g \Rightarrow x = 0.25(L-x)$$

$$x = \frac{L}{4} - \frac{x}{4} \Rightarrow x + \frac{x}{4} = \frac{L}{4} \Rightarrow x = 0.25(L-x)$$

The maximum fraction hanging is

$$= \frac{x}{L} = \frac{1}{5} = 20\%.$$

(NCERT Ed. 2025-26, 11th, Page No. 60)

2. (2)

From Stefan's Law and calorimetry, the rate of cooling of a body is given by:

$$\frac{dT}{dt} \propto \frac{Surface\ Area\,(A)}{m.c} \propto \frac{A}{V.\rho.c}.$$

$$\frac{R_s}{R_c} = \frac{A_s}{A_c}.$$

We must express A_s and A_c purely in terms of volume V.

For the sphere: $V = \frac{4}{3}\pi R^3 \Rightarrow R = \left(\frac{3V}{4\pi}\right)^{1/3}.$

Surface area

$$A_s = 4\pi R^3 = 4\pi\left(\frac{3V}{4\pi}\right)^{2/3}$$

$$= (4\pi)^{1/3}(3V)^{2/3} = \left(36\pi V^2\right)^{1/3}$$

For the cube:

$$V = 6a^2 = 6\left(V^{1/3}\right)^3 = 6V^{2/3} = \left(216V^2\right)^{1/3}$$

The ratio of their cooling rates is:

$$\frac{R_s}{R_c} = \frac{\left(36\pi V^2\right)^{1/3}}{\left(216V^2\right)^{1/3}} = \left(\frac{36\pi}{216}\right)^{1/3} = \left(\frac{\pi}{6}\right)^{1/3}$$

(NCERT Ed. 2025-26, 11th, Page No. 220)

3. (4)

$$P = V_z.I_{max} \Rightarrow I_{max} = \frac{P}{V_z} = \frac{0.5\text{W}}{10\text{V}} = 0.05\text{A}.$$

When the load resistor is disconnected, all the current from the supply flows directly through the Zener diode. To prevent damage, this current must not exceed $I_{max} = 0.05$ A.

The voltage drop across the series resistor R_s is:

$$V_s = V_{supply} - V_2 = 20\text{V} - 10\text{V} = 10\text{V}.$$

Using Ohm's law to find the minimum series resistance R_s:

$$R_s = \frac{V_s}{I_{max}} = \frac{10\text{V}}{0.05\text{A}} = 200\Omega$$

(NCERT Ed. 2025-26, 12th, Page No. 336)

4. (3)

The process is P $\propto$ V $\Rightarrow PV^{-1}$ = constant. This a polytropic process with index x = –1.

Work done $W = \frac{nR\Delta T}{1-x} = \frac{1.R\Delta T}{1-(-1)} = \frac{R\Delta T}{2}$

The molar heat capacity is

$$C = C_v + \frac{R}{1-x} = \frac{3R}{2} + \frac{R}{2} = 2R$$

Heat supplied $\Delta Q = nC\Delta T = 1.2R.\Delta T = 2R\Delta T.$

The fraction of heat converted into work is

$$\frac{W}{\Delta Q} = \frac{R\Delta T/2}{2R\Delta T} = \frac{1}{4}.$$

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5. (1)

The power dissipated in the external resistor is given $P = \left(\frac{E}{R+r}\right)^2 R.$

It is a standard property that if power is same for two different resistance R_1 and R_2, the internal resistance of the cell is the geometric mean of the two:

$r = \sqrt{R_1 R_2} = \sqrt{2\times 8} = \sqrt{16} = 4\Omega.$

Maximum power theorem states that maximum power is drawn when $R = r$. The maximum power is given by:

$$P_{max} = \frac{E^2}{4r}$$

When are given P_{max} = 9W and we found r = 4Ω

$$9 = \frac{E^2}{4\times 4} \Rightarrow E^2 = 144 \Rightarrow E = 12\text{V}.$$

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6. (4)

Power P = F. $v = \left(m\frac{dv}{dt}\right)v.$

Separating variable and integrating:

$\int_0^v mudu = \int_0^t P\,dt \Rightarrow \frac{1}{2}mu^2 = \mathrm{P}t \Rightarrow u = \sqrt{\frac{2P}{m}}t^{1/2}.$

Since $u = \frac{dx}{dt}$, we integrate again to find displacement x:

$x = \int\sqrt{\frac{2P}{m}}t^{1/2}dt = \sqrt{\frac{2P}{m}}\left(\frac{2}{3}t^{3/2}\right) \Rightarrow x \propto t^{3/2}$

Distance covered in first t_0 second: $x_1 = k(t_0)^{3/2}$.

Distance covered up to $4t_0$: $x_{total} = k(4t_0)^{3/2} = k.\ 8(t_0)^{3/2} = 8x_1$

Distance covered in the interval t_0 to $4t_0$: $x_2 = x_{total} - x_1 = 8x_1 - x_1 = 7x_1$

Therefore, the ratio $x_1 : x_2 = 1:7$.

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7. (3)

Consider a small element of length dx at a distance x from the free (bottom) end of the rope.

The tension T at this cross-section due to the weight of the rope portion below it (length x).

Mass of this portion is $m(x) = \left(\frac{M}{L}\right)x.$

Tension $T = m(x)g = \left(\frac{M}{L}\right)xg.$

The elastic potential energy dU stored in this small element dx is given by:

$dU = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$

$= \frac{1}{2}\left(\frac{T}{A}\right)\left(\frac{T}{AV}\right)(Adx) = \frac{T^2dx}{2AY}$

Substitute T:

$dU = \frac{1}{2AY}\left(\frac{Mgx}{L}\right)^2 dx = \frac{M^2g^2}{2AYL^2}x^2dx.$

Integrate from $x = 0$ to $x = L$ to find the total energy:

$U = \int_0^L \frac{M^2g^2}{2AYL^2}x^2dx = \frac{M^2g^2}{2AYL^2}\left[\frac{x^3}{3}\right]_0^L$

$= \frac{M^2g^2}{2AYL^2}\left(\frac{L^3}{3}\right) = \frac{M^2g^2L}{6AY}$

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8. (3)

Given the angle of deviation is $\theta = 60^o = \frac{\pi}{3}$ radians.

The time period for a complete circle (360º or 2π radians) is $T = \frac{2\pi m}{qB}$.

The time t spent in the field covering an angle θ is:

$t = \frac{\theta}{2\pi} \times T = \frac{\pi/3}{2\pi} \times \left(\frac{2\pi m}{qB}\right) = \frac{1}{6} \times \frac{2\pi m}{qB} = \frac{\pi m}{3qB}$

(NCERT Ed. 2025-26, 12th, Page No. 112)

9. (1)

For Helium (monoatomic): $n_1 = 2,\ C_{v1} = \frac{3}{2}R$

For Helium (diatomic): $n_2 = 3,\ C_{v2} = \frac{5}{2}R$

The equivalent molar heat capacity at constant volume (C_v, mix) for the mixture is:

$C_v, mix = \frac{m_1C_{v_1+n_2}C_{v2}}{n_1+n_2} = \frac{2\left(\frac{3}{2}R\right)+3\left(\frac{5}{2}R\right)}{2+3}$

$= \frac{3R+7.5R}{5} = \frac{10.5R}{5} = 2.1R$

By Mayer's relation, the equivalent specific heat at constant pressure $\left(C_{p,mix}\right)$ is:

$C_{P,mix} = C_{v,mix} + R = 2.1R + R = 3.1R$

The ratio of specific heat γ_{mix} is:

$\gamma_{mix} = \frac{C_{P,mix}}{C_{v,mix}} = \frac{3.1R}{2.1R} = \frac{31}{21} \approx 1.476.$

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10. (4)

The orbital radius from the center of the Earth is $r = R + h = R + 2R = 3R$.

The total mechanical energy of a satellite in a circular orbit of radius r is $E = -\frac{GMm}{2r}$

substituting r = 3R, the current energy of the satellite is $E_i = -\frac{GMm}{6R}$.

To escape to infinity, the final total energy must be at least $E_f = 0$.

The required extra (Binding Energy) is:

$\Delta E = E_f - E_i = 0 - \left(-\frac{GMm}{6R}\right) = \frac{GMm}{6r}.$

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11. (2)

For an equilateral prism, the refracting angle A = 60°

Symmetrical passage implies the prism is the minimum deviation position $(\delta = \delta_m)$.

The question states that the angle of minimum deviation $\delta_m = A = 60^\circ$.

Using the prism formula for refractive index μ

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{\sin\left(\frac{60^\circ+60^\circ}{2}\right)}{\sin\left(\frac{60^\circ}{2}\right)} = \frac{\sin(60^\circ)}{\sin(30^\circ)}.$$

$$\mu = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$$

(NCERT Ed. 2025-26, 12th, Page No. 240)

12. (3)

Acceleration due to gravity is purely downwards:

$\vec{a} = -10\hat{j}$

Instantaneous velocity at any time *t* is

$\vec{v}(t) = \vec{u} + \vec{a}t = 3\hat{i} + (4 - 10t)\hat{j}.$

For the vectors to be perpendicular, their dot product must be zero:

$\vec{u}.\vec{v}(t) = 0$

$(3\hat{i} + 4\hat{j})(3\hat{i} + (4-10t)\hat{i}) = 0$

$3(3) + 4(4 - 10t) = 0$

$9 + 16 - 40t = 0 \Rightarrow 25 = 40t$

$\Rightarrow t = \frac{25}{40} = \frac{5}{8} = 0.625\ s$

(NCERT Ed. 2025-26, 11th, Page No. 39)

13. (3)

The magnetic field lines generated by the outer coil pass parallel to the surface of the inner coil (perpendicular to its area vector). Therefore, the magnetic flux $\phi = \vec{B}.\vec{A} = 0$, making the mutual inductance zero.

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14. (2)

Total initial mechanical energy:

$$E_i = \frac{1}{2}mu^2 = \frac{1}{2}m(4gL) = 2mgL.$$

When the string is horizontal, the bob has risen by a height h = L.

Potential energy at this point: $U_f = mgL$.

By conservation of mechanical energy ($E_i = E_f$):

$$2mgL = mgL + \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2}mv^2 = mgL \Rightarrow v^2 = 2gL.$$

When the string is perfectly horizontal, gravity acts purely downwards (tangential to the path), so it contributes nothing to the radial centripetal force. The only horizontal radial force is the tension *T*.

$$T = \frac{mv^2}{L} = \frac{m(2gL)}{L} = 2mg.$$

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15. (1)

When connected in parallel with an identical uncharged capacitor (C), the total charge Q = CV is shared equality because the capacitances are equal.

The common potential becomes

$$V_{common} = \frac{Total\ Charge}{Total\ Capacitance} = \frac{CV}{C+V} = \frac{V}{2}$$

The final total energy stored is:

$$U_f = \frac{1}{2}C_{eq}(V_{common})^2$$

$$= \frac{1}{2}(2C)\left(\frac{V}{2}\right)^2 = \frac{1}{2}(2C)\left(\frac{V^2}{4}\right) = \frac{1}{4}CV^2.$$

The ratio of final to initial energy is:

$$\frac{U_f}{U_i} = \frac{\frac{1}{4}CV^2}{\frac{1}{2}CV^2} = \frac{1}{4} \times \frac{2}{1} = \frac{1}{2}$$

(NCERT Ed. 2025-26, 12th, Page No. 75)

16. (2)

The net RMS voltage of the source in a series LCR circuit is given by phasors:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{20^2 + (40-40)^2} = \sqrt{20^2 + 0} = 20V$$

Since $V_L = V_C$, the circuit is in a state of electrical resonance.

At resonance, the impedance is purely resistive (Z = R), and the voltage and current are in phase.

Power factor $\cos\phi = \frac{V_R}{V} = \frac{R}{Z}$.

$\cos\phi = \frac{20}{20} = 1.0.$

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17. (2)

Closed pipe (in water): 1st overtone is the 3rd harmonic.

$f_{closed} = 3\left(\frac{v}{4L}\right) = 450\text{Hz} \Rightarrow \frac{v}{4L} = 150\text{Hz}.$

Open pipe (out of water): Fundamental frequency.

$f_{open} = \frac{v}{2L} = 2 \times 150 = 300\,Hz$

(NCERT Ed. 2025-26, 11th, Page No. 292)

18. (3)

Time period is given by:

$= 2\pi\sqrt{\frac{L}{g}} \Rightarrow g = 4\pi^2 \frac{L}{T^2}.$

The fractional error equation is:

$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\left(\frac{\Delta T}{T}\right)$

Given:

$\frac{\Delta L}{L} = \frac{0.1}{100} = 0.001$

$\frac{\Delta T}{T} = \frac{0.01}{2} = 0.005$

Total relative error.

$\frac{\Delta g}{g} = 0.001 + 2(0.005) = 0.001 + 0.010 = 0.011$

Percentage error = 0.011 × 100% = 1.1%.

(NCERT Ed. 2025-26, 11th, Page No. 7)

19. (1)

Objective Lens analysis.

$u_0 = -2.5cm, f_0 = +2.0cm$

Using lens formula $\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$:

$\frac{1}{v_0} - \frac{1}{-2.5} = \frac{1}{2} \Rightarrow \frac{1}{v_0} + \frac{2}{5} = \frac{1}{2}$

$\frac{1}{v_0} = 0.5 - 0.4 = 0.1 \Rightarrow v_0 = 10$

Magnification of objective

$\frac{v_0}{u_0} = \frac{10}{-2.5} = -4.$

Magnification $|m_0| = 4$

Eyepiece analysis.

The eyepiece acts as a simple magnifier forming the image at D.

Magnification of eyepiece $m_e = 1 + \frac{D}{f_e}$

$m_e = 1 + \frac{25}{5} = 1 + 5 = 6$

Step 3: Total Magnification.

$M = |m_0| \times m_e = 4 \times 6 = 24$

(NCERT Ed. 2025-26, 12th, Page No. 244)

20. (3)

The magnetic Force $\left(\vec{F} = q\left(\vec{v} \times \vec{B}\right)\right)$ always acts perpendicular to the velocity of the particle.

Because the force is perpendicular to motion, it does no work on the electron. Therefore, the kinetic energy and speed (v) remain constant (only the direction changes).

Conclusion: Since v is constant, the wavelength λ remains constant.

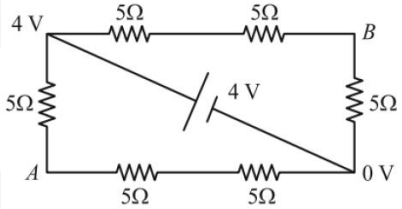
(NCERT Ed. 2025-26, 12th, Page No. 112)

21. (2)

KTG postulates that all collisions (molecule-molecule-wall) are perfectly elastic, meaning there is no loss of kinetic energy during impact.

(NCERT Ed. 2025-26, 11th, Page No. 249)

22. (3)



$V_A = \frac{8}{3}\text{V}$ and $V_B = \frac{4}{3}\text{V}$

$V_A - V_B = \frac{4}{3}\text{V}.$

(NCERT Ed. 2025-26, 12th, Page No. 97)

23. (1)

Electrostatic forces are conservative. Work done by these forces is independent of path.

(NCERT Ed. 2025-26, 12th, Page No. 46)

24. (2)

For equilibrium:

$qE = \text{mg}$

$E = \frac{mg}{q} = \frac{10^{-6} \times 10}{10^{-6}} = 10V/m.$

As the charge is negative, electric field should be vertically downwards.

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25. (3)

Net potential of grounded sphere = 0

$$\frac{kq'}{r_1}+\frac{kq}{r_2}=0.$$

$$q'=-q\frac{r_1}{r_3}$$

(NCERT Ed. 2025-26, 12th, Page No. 62)

26. (4)

$$U=\frac{1}{2}kx^2.At\,x=A/2,$$

$$U=\frac{1}{2}K\left(A/2\right)^2=\frac{1}{4}\left(\frac{1}{2}kA^2\right).$$

Total Energy E $=\frac{1}{2}kA^2.soU=E/4$

$$2.5=E/4\Rightarrow E=10J..$$

(NCERT Ed. 2025-26, 11th, Page No. 268)

27. (2)

Volume conservation:

$$2\left(\frac{4}{3}\pi R^3\right)=\frac{4}{3}\pi R'^3\Rightarrow R'=2^{1/3}R.$$

Surface Area ratio

$$A_i/A_f=\frac{2\left(4\pi R^2\right)}{4\pi\left(2^{1/3}R\right)^2}=\frac{2}{2^{2/3}}=2^{1/3}$$

Energy $\propto$ Area.

(NCERT Ed. 2025-26, 11th, Page No. 193)

28. (1)

$$\omega_g^2=\omega_i^2+2\alpha\theta$$

$$0=\left(\frac{\pi}{10}\right)^2+2\alpha\left(4\pi^2\right)$$

$$\alpha=\frac{-1}{800}\text{ rad/s}^2$$

$\tau = I\propto = -2\times10^{-6}$ Nm.

(NCERT Ed. 2025-26, 11th, Page No. 119)

29. (3)

For no slipping, the pseudo force ma balances mg sin θ along the incline, but we need Normal Force.

Vertical equilibrium: N cos θ = mg $\Rightarrow$ N $=\frac{mg}{\cos\theta}$

(NCERT Ed. 2025-26, 11th, Page No. 65)

30. (3)

$$B_{axis}=\frac{\mu_0IR^2}{2\left(R^2+x^2\right)^{3/2}}.B_{center}=\frac{\mu_0I}{2R}$$

$$\frac{1}{\left(R^2+x^2\right)^{3/2}}=\frac{1}{8R^3}$$

Solving gives x = $\sqrt{3}R$.

(NCERT Ed. 2025-26, 12th, Page No. 116)

31. (2)

Lyman longest (2 to 1): $\frac{1}{\lambda_L}=R\left(1-\frac{1}{4}\right)=3R/4$

Balmer longest (3 to 2): $R\left(\frac{1}{4}-\frac{1}{9}\right)=5R/36$

$$\frac{\lambda_B}{\lambda_L}=\frac{3R/4}{5R/36}=\frac{3}{4}\times\frac{36}{5}=\frac{27}{5}$$

(NCERT Ed. 2025-26, 12th, Page No. 297)

32. (4)

- Diamagnetic materials have a small negative magnetic susceptibility ($\chi < 0$)
- Superconductors exhibit the Meissner effect and behave as perfect diamagnets, having magnetic susceptibility of $\chi = -1$ (which is also negative)
- Therefore, susceptibility is negative for both.

(NCERT Ed. 2025-26, 12th, Page No. 147)

33. (3)

Current in discharging LR circuit at time t is,

$$i=i_0r^{-t/\tau}$$

$$\Rightarrow t=\frac{L}{R}\ln\frac{100}{37}=\frac{L}{R}$$

(NCERT Ed. 2025-26, 12th, Page No. 168)

34. (3)

Frequency order is

$f_{\gamma\text{ rays}} > f_{x\text{ rays}} > f_{\text{micro wave}} > f_{\text{radio wave}}$

(NCERT Ed. 2025-26, 12th, Page No. 209)

35. (2)

B.E.= $[ZM_p + (A - Z)M_n - M(A, Z)]c^2$

$$\therefore\frac{B.E.}{c^2}=ZM_p+\left(A-Z\right)M_n-M\left(A,Z\right)$$

or $M(A, Z) = ZM_p + (A - Z)\,M_n - B.E./c^2$

(NCERT Ed. 2025-26, 12th, Page No. 311)

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36. (3)

In a denser medium, the speed of light (v) decreases. According to Huygens' Principle, secondary wavelets travel a shorter distance (d = v × t) in the same time interval compared to a rarer medium. This difference in distance causes the new wavefront (the tangent) to tilt, resulting in the light bending towards the normal. Thus, the Reason correctly explains the Assertion.

(NCERT Ed. 2025-26, 12th, Page No. 260)

37. (3)

$$\beta = \frac{\lambda D}{d} = \frac{600\times10^{-9}\times1}{2+10^{-3}} = 300\times10^{-16}\times10^{3}$$

= 0.3 mm

(NCERT Ed. 2025-26, 12th, Page No. 266)

38. (3)

Given, $\frac{A_1}{A_2} = \frac{4}{3}$

$$\therefore \sqrt{\frac{I_1}{I_2}} = \frac{4}{3} \qquad (\text{as } I \propto A^2)$$

$$\frac{I_{max}}{I_{max}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2}$$

(NCERT Ed. 2025-26, 12th, Page No. 265)

39. (3)

For completing the circle V:

$v \geq \sqrt{5gR}$

$v \geq \sqrt{2gR} \Rightarrow H \geq \frac{5}{2}R$

Or $R \leq \frac{2\times H}{5}$

Maximum value of R = 2 cm.

If R ≤ 2 cm, the body completes the circle. If R > 2 cm, it will not complete the circle.

(NCERT Ed. 2025-26, 11th, Page No. 79)

40. (1)

$v_m = \sqrt{v_1^2 + v_2^2}$

$v_m = 5m/s, v_1 = 1m/s$

$25 = \frac{(1)^2 + v_2^2}{2}$

$50 - 1 = v_2^2$

v_2 = 7 m/s

(NCERT Ed. 2025-26, 11th, Page No. 18)

41. (1)

As $I_1 = I_2$

$$\frac{1}{4}MR_1^2 = \frac{2}{3}MR_2^2$$

$$\therefore \frac{R_1}{R_2} = \sqrt{\frac{8}{3}}$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

42. (2)

Gravitational force is always attractive and a long-range force. It is independent of the presence of other particles.

(NCERT Ed. 2025-26, 11th, Page No. 139)

43. (1)

$$W = \int_{V_0}^{2V_0} P_0\left(1 + \frac{V^2}{V_0^2}\right)dV$$

$$W = P_0\left[V + \frac{V^3}{3V_0^2}\right]_{V_0}^{2V_0}$$

$$W = P_0\left[(2V_0 - V_0) + \frac{\left(8V_0^3 - V_0^3\right)}{3V_0^2}\right]$$

$$W = P_0\left[V_0 + \frac{7V_0}{3}\right] = \frac{10}{3}P_0V_0$$

(NCERT Ed. 2025-26, 11th, Page No. 230)

44. (3)

A wave travelling in positive x-direction may be represented as $y = A\sin\frac{2\pi}{\lambda}(vt - x)$. On putting values

$$y = 0.2\sin\frac{2\pi}{60}(360t - x)$$

$$\Rightarrow y = 0.2\sin 2\pi\left(6t - \frac{x}{60}\right)$$

(NCERT Ed. 2025-26, 11th, Page No. 282)

45. (4)

The core of a transformer is laminated to reduce power leakage due to eddy current.

(NCERT Ed. 2025-26, 12th, Page No. 196)

[Contd...

46. (4)

More the charge on cation (or) anion, more will be lattice energy of ionic compound.

(NCERT Ed. 2025-26 11th Part-I Page No. 112)

47. (1)

$$SO_2 + Ca(OH)_2 \rightarrow \underset{\text{milky}}{CaSO_3} + H_2O$$

48. (3)

$$-I \text{ effect} : -\overset{+}{N}H_3 > -CHO > -Br$$

(NCERT Ed. 2025-26 11th Part-II Page No. 274)

49. (4)

$X_{H_2SO_4} = 0.1 \qquad \therefore\ n_{H_2SO_4} = 0.1$

$N_{H_2O} = 0.9$

$W_{H_2O} = 0.9 \times 18\ g$

$$\text{Molality} = \frac{0.1}{0.9 \times 18 \times 10^{-3}} = 6.17\ m$$

(NCERT Ed. 2025-26 12th Part-I Page No. 05)

50. (3)

- Diamond: sp^3
- Buckminsterfullerene: sp^2

51. (3)

$M^{2+} + e^- \rightarrow M^+, \quad E_1^o = x\ V$

$M^+ + e^- \rightarrow M, \quad E_2^o = y\ V$

$M^{2+} + 2e^- \rightarrow M, \quad E_3^o$

Now, $\Delta G_3^o = \Delta G_1^o + \Delta G_2^o$

$$E_3^o = \frac{n_1E_1^o + n_2E_2^o}{n_3} = \frac{x+y}{2} V$$

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

52. (2)

- $[PtCl_4]^{2-} : dsp^2$ hybridise
- $[Ni(CN)_4]^{2-} : dsp^2$ hybridise

(NCERT Ed. 2025-26 12th Part-I Page No. 130)

53. (1)

$\Delta n_g = 1 - 2 = -1$

$$K_p = K_c(RT)^{-1} = \frac{K_c}{RT}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 178)

54. (3)

Curdling of milk is caused due to the formation of lactic acid by the bacteria present in milk.

(NCERT Ed. 2025-26 12th Part-II Page No. 295)

55. (2)

- Os: 5d series
- Ac: 6d series
- Sc: 3d series
- Ru: 4d series

(NCERT Ed. 2025-26 12th Part-I Page No. 91)

56. (2)

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

(NCERT Ed. 2025-26 11th Part-II Page No. 247)

57. (2)

$|m_\ell| = 2$ means $m_\ell = +2$ or -2

$\therefore$ 4d : $\overset{m_\ell}{\boxed{-2}\boxed{-1}\boxed{0}\boxed{+1}\boxed{+2}}$

4f : $\overset{m_\ell}{\boxed{-3}\boxed{-2}\boxed{-1}\boxed{0}\boxed{+1}\boxed{+2}\boxed{+3}}$

For $m_s = +\frac{1}{2}$, only 4 electron are possible.

(NCERT Ed. 2025-26 11th Part-I Page No. 63)

58. (2)

Binary mixture of C_6H_5OH and $C_6H_5NH_2$ shows negative deviation from Raoult's law. Vapour pressure of solution is less than vapour pressure of pure C_6H_5OH and $C_6H_5NH_2$, so the boiling point of solution is more than boiling point of pure phenol and aniline, hence it form maximum boiling azeotrope.

(NCERT Ed. 2025-26 12th Part-I Page No. 14)

59. (1)

- Due to more close packing, melting point of neo-pentane is more than that of n-pentane.
- Due to more surface area, boiling point of n-pentane is more than that of neo-pentane.

(NCERT Ed. 2025-26 11th Part-II Page No. 302)

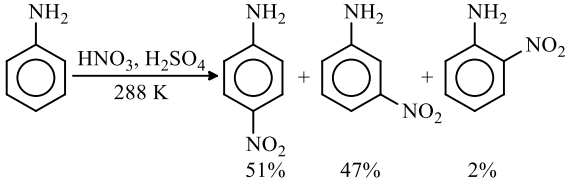
60. (1)

Metallic character increases down a group and decreases along a period as we move from left to right. Hence, the order of increasing metallic character is: Be < Mg < Na

(NCERT Ed. 2025-26 11th Part-I Page No. 94)

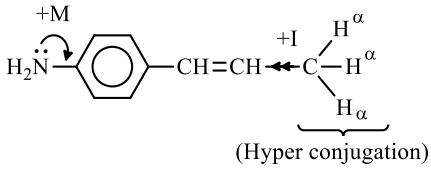
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61. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 273)

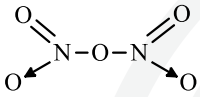
62. (1)



(NCERT Ed. 2025-26 11th Part-II Page No. 277)

63. (2)

- In solid state PCl_5 exist as an ionic solid, $[PCl_4]^+[PCl_6]^-$.
- N_2O_5:



64. (4)

$X_2 \rightleftharpoons 2X, K_{eq} = \frac{[X]^2}{[X_2]}$

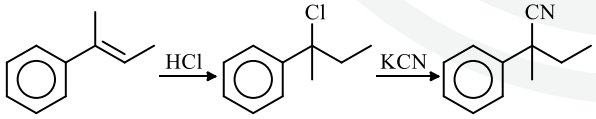
$\therefore [X]^2 = K_{eq}[X_2]$

$\text{rate} = k[X]^2[Y] = kK_{eq}[X_2][Y]$

$\therefore$ overall order = 1 + 1 = 2

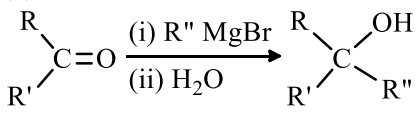
(NCERT Ed. 2025-26 12th Part-I Page No. 68)

65. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 170)

66. (4)



(NCERT Ed. 2025-26 12th Part-II Page No. 201)

67. (2)

Among 3d series, enthalpy atomisation is lowest and highest respectively for Zn and V.

(NCERT Ed. 2025-26 12th Part-I Page No. 94)

68. (2)

$\mu_{100\% \text{ Ionic}} = q \times d = 1.6 \times 10^{-19} \times 2.6 \times 10^{-10}$

$= 4.16 \times 10^{-29}$ C m

$\%$ ionic character $= \frac{3.336 \times 10^{-29}}{4.16 \times 10^{-29}} \times 100 = 80.2\%$

(NCERT Ed. 2025-26 11th Part-I Page No. 111)

69. (1)

Boron has high melting point.

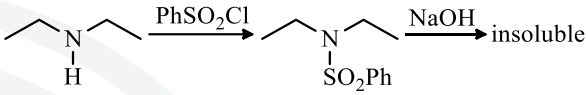
70. (2)

$W = -P_{ext}(V_2 - V_1) = -2(20 - 10) = -20 \text{ L} - \text{atm}$

$= -20 \times 101.3 \text{ J} = -2026 \text{ J} = -2.02 \text{ kJ}$

(NCERT Ed. 2025-26 11th Part-I Page No. 142)

71. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 271)

72. (1)

- Combination reaction: $C + O_2 \rightarrow CO_2$
- Non-redox reaction: $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
- Metal displacement redox reaction: $CuSO_4 + Zn \rightarrow Cu + ZnSO_4$
- Non-metal displacement redox reaction: $2Na + 2H_2O \rightarrow 2NaOH + H_2$

(NCERT Ed. 2025-26 11th Part-II Page No. 242)

73. (3)

$A(g) + 6B(g) \rightleftharpoons 4C(g)$

(i)	1	1	–
(f)	$1-x$	$1-6x$	$4x$

$\therefore$ at eq., $[A] > [B]$

(NCERT Ed. 2025-26 11th Part-I Page No. 183)

74. (2)

- Glutamic acid: Acidic amino acid
- Glycine: Optically inactive amino acid
- Cysteine: Sulphur containing amino acid
- Lysine: Basic amino acid

(NCERT Ed. 2025-26 12th Part-II Page No. 291)

75. (2)

C_2H_5ONa is a strong nucleophile as well as a strong base. Thus elimination reaction predominates over substitution as major product will be 2-methylprop-1-ene.

(NCERT Ed. 2025-26 12th Part-II Page No. 216)

[Contd...

76. (3)

Higher the value of K_H at given pressure, lower the solubility of gas in liquids.

(NCERT Ed. 2025-26 12th Part-I Page No. 07)

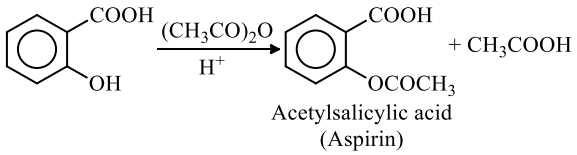
77. (2)

Energy of orbitals for H-atom:

$1s < 2s = 2p < 3s = 3p = 3d$.

(NCERT Ed. 2025-26 11th Part-I Page No. 60)

78. (3)

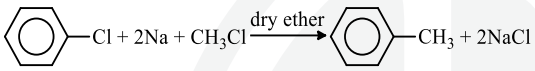


Acetylsalicylic acid
(Aspirin)

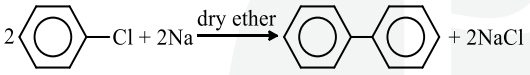
(NCERT Ed. 2025-26 12th Part-II Page No. 208)

79. (1)

- Wurtz fittig reaction:



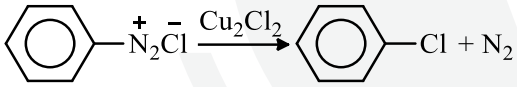
- Fittig reaction:



- Finkelstein reaction:

$CH_3CH_2Cl + NaI \rightarrow CH_3CH_2I + NaCl$

- Sandmeyer's reaction:



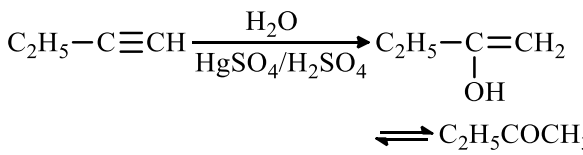
(NCERT Ed. 2025-26 12th Part-II Page No. 186)

80. (2)

- $2.24\ L\ O_2 : \frac{2.24}{22.4} \times 32 = 3.2\ g$
- $6.022 \times 10^{21}\ molecule : \frac{6.022 \times 10^{21}}{6.022 \times 10^{23}} \times 32 = 0.32\ g$
- 1 g molecule: 1 mole O_2 = 32 g
- 0.001 mol: 0.001 × 32 = 0.032 g

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

81. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 317)

82. (1)

$BP : \underset{\text{(H-bonding)}}{SbH_3} > NH_3 > AsH_3 > PH_3$

83. (4)

$pH : \underset{(pH>7)}{CH_3COONa} > \underset{(pH=7)}{NaCl} > \underset{(pH<7)}{NH_4Cl} > \underset{(pH<<7)}{HCl}$

(NCERT Ed. 2025-26 11th Part-I Page No. 202)

84. (4)

- Brown ring test is used for the conformation of NO_3^- ion.
- The colour of lead sulphide precipitate is black.

85. (4)

$$\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{4 \times 10^{-5} \times 1000}{0.001} = 40\ S\ cm^2\ mol^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{40}{400} = 0.1$$

(NCERT Ed. 2025-26 12th Part-I Page No. 50)

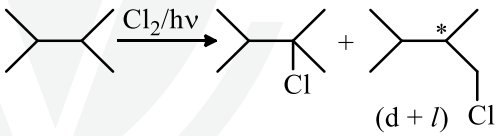
86. (3)

$$+\frac{d[SO_3]}{dt} = -\frac{d[SO_2]}{dt} = \frac{32}{64}\ mol\ s^{-1} = 0.5\ mol\ s^{-1}$$

$$\therefore \frac{d[SO_3]}{dt} = 0.5 \times 80\ g\ s^{-1} = 40\ g\ s^{-1}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 65)

87. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 165)

88. (4)

$COCl_3 \cdot 6NH_3 : [Co(NH_3)_6]Cl_3 : 4\ ions$

(NCERT Ed. 2025-26 12th Part-I Page No. 120)

89. (1)

Mayer's relation: $(C_p)_m - (C_v)_m = R$

(NCERT Ed. 2025-26 11th Part-I Page No. 145)

90. (2)

Atomic radii decrease across a period. Cations are smaller than their parent atoms. Among isoelectronic species, the one with the larger positive nuclear charge will have a smaller radius. Hence, the largest species is Mg and the smallest one is Al^{3+}.

(NCERT Ed. 2025-26 11th Part-I Page No. 87)

[Contd...

91. (2)

The function of eukaryotic mitochondria is performed by mesosomes in prokaryotes.

(NCERT Ed. 2025-26, 11th, Page No. 90)

92. (4)

Eucalyptus is an Angiosperm (flowering plant). In angiosperms, seeds are enclosed within a fruit. *Sequoia* is a Gymnosperm. Gymnosperms are characterized by having naked seeds. *Salvinia* is a heterosporous Pteridophyte. The development of heterospory (producing two types of spores: macro and micro) and the retention of the female gametophyte on the parent sporophyte are considered the evolutionary precursors to the seed habit. *Sphagnum* is a moss (bryophyte). In all bryophytes, the dominant, independent, and photosynthetic phase of the life cycle is the haploid gametophyte.

(NCERT Ed. 2025-26, 11th, Page No. 29, 32, 34)

93. (3)

In bacteria, since the mRNA does not require any processing to become active, and also since transcription and translation take place in the same compartment (there is no separation of cytosol and nucleus in bacteria), many times the translation can begin much before the mRNA is fully transcribed. Consequently, the transcription and translation can be coupled in bacteria.

(NCERT Ed. 2025-26, 12th, Page No. 94)

94. (2)

The innermost layer of the cortex in dicot root is called endodermis. It comprises a single layer of barrel-shaped cells without any intercellular spaces. Pericycle of dicot stem is usually composed of semi-lunar patches of sclerenchyma. Hypodermis of dicot stem consists of a few layers of collenchymatous cells (oval, spherical or polygonal). Palisade parenchyma cells of dorsiventral leaf are typically elongated.

(NCERT Ed. 2025-26, 11th, Page No. 74)

95. (4)

The physico-chemical approach to study and understand living organisms is called 'Reductionist Biology'. It can also explain the abnormal processes that occur during any diseased condition.

(NCERT Ed. 2025-26, 11th, Page No. 87)

96. (1)

In a typical angiosperm, the sequence of formation is:

Megaspore (A) – formed by meiosis of the megaspore mother cell

Embryo sac (D) – develops from the functional megaspore

Endosperm (C) – formed first after double fertilization

Embryo (B) – develops from the zygote usually after formation of endosperm

Fruit (E) – ovary develops into fruit after fertilization

(NCERT Ed. 2025-26, 12th, Page No. 10, 11, 18, 19, 20)

97. (4)

Since splitting of the water molecule takes place on the inner side of the thylakoid membrane, the protons or hydrogen ions that are produced by the splitting of water accumulate within the lumen of the thylakoids. During chemiosmosis, the primary accepter of electron which is located towards the outer side of the membrane transfers its electron not to an electron carrier but to an H carrier. Hence, this molecule removes a proton from the stroma while transporting an electron. When this molecule passes on its electron to the electron carrier on the inner side of the membrane, the proton is released into the inner side or the lumen side of the membrane. This leads to reduction of pH in thylakoid lumen.

(NCERT Ed. 2025-26, 11th, Page No. 141)

98. (2)

The given cross is TtRr × ttrr. The gametes from TtRr are TR, Tr, tR, tr. The gamete from ttrr is tr. The progeny will be TtRr, Ttrr, ttRr, ttrr in a 1:1:1:1 ratio. The recessive parent phenotype is dwarf and wrinkled (ttrr), which constitutes 1/4 of the progeny.

(NCERT Ed. 2025-26, 12th, Page No. 64)

99. (3)

Cells in G_0 phase remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.

(NCERT Ed. 2025-26, 11th, Page No. 122)

100. (4)

Some common herbivores are insects, birds and mammals in terrestrial ecosystem and molluscs and zooplankton in aquatic ecosystem.

(NCERT Ed. 2025-26, 12th, Page No. 209)

[Contd...

101. (1)

Trypanosoma is a flagellated protozoan that causes sleeping sickness in humans. *Paramoecium* is a ciliated protozoan. It possesses thousands of hair-like cilia on its surface that beat in a coordinated manner, allowing the organism to move actively and steer food into its gullet. *Plasmodium* is classified as a sporozoan. It is characterized by having an infectious, spore-like stage (sporozoites) in its life cycle. It is the causative agent of malaria. *Entamoeba* is an amoeboid protozoan. It moves and captures prey using pseudopodia (false feet) and is parasitic.

(NCERT Ed. 2025-26, 11th, Page No. 15, 16)

102. (3)

In general, species diversity decreases as we move away from the equator towards the poles. A forest in a tropical region like Equador has up to 10 times as many species of vascular plants as a forest of equal area in a temperate region like the Midwest of the USA.

(NCERT Ed. 2025-26, 12th, Page No. 219)

103. (2)

Poaceae is a monocot plant family.

(NCERT Ed. 2025-26, 11th, Page No. 8)

104. (4)

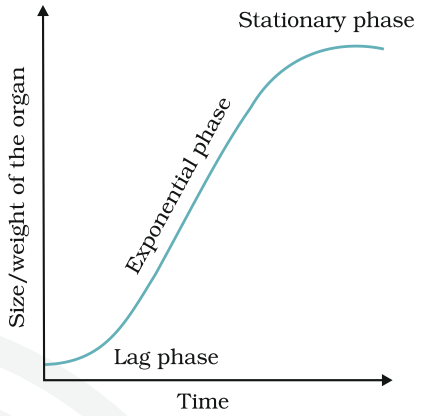
Lactic Acid Bacteria improve the nutritional quality of milk by increasing vitamin B_{12}. The dough, which is used for making foods such as *dosa* and *idli* is fermented by bacteria.

(NCERT Ed. 2025-26, 12th, Page No. 151)

105. (3)

In arithmetic growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures. The simplest expression of arithmetic growth is exemplified by a root elongating at a constant rate. On plotting the length of the organ against time, a linear curve is obtained. Mathematically, it is expressed as $L_t = L_0 + rt$.

In most systems, the initial growth is slow (lag phase), and it increases rapidly thereafter – at an exponential rate (log or exponential phase). Here, both the progeny cells following mitotic cell division retain the ability to divide and continue to do so. However, with limited nutrient supply, the growth slows down leading to a stationary phase. If we plot the parameter of growth against time, we get a typical sigmoid or S-curve. A sigmoid curve is a characteristic of living organism growing in a natural environment. The exponential growth can be expressed as $W_1 = W_0 e^{rt}$.



An idealised sigmoid growth curve typical of cells in culture, and many higher plants and plant organs

(NCERT Ed. 2025-26, 11th, Page No. 170)

106. (1)

There are instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. An example of this is the disease phenylketonuria, which occurs in humans. The disease is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). This inborn error of metabolism is inherited as the autosomal recessive trait. The affected individual lacks an enzyme that converts the amino acid phenylalanine into tyrosine. This manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation.

(NCERT Ed. 2025-26, 12th, Page No. 75)

107. (1)

Fermentation accounts for only a partial breakdown of glucose whereas in aerobic respiration it is completely degraded to CO_2 and H_2O. In fermentation there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid whereas many more molecules of ATP are generated under aerobic conditions.

(NCERT Ed. 2025-26, 11th, Page No. 157)

[Contd...

108. (2)

A dicotyledonous stem has open vascular bundles with endarch primary xylem. The xylem and phloem are located on the same radius (called conjoint). Radial arrangement (where xylem and phloem alternate on different radii) is a characteristic of roots. In the conjoint vascular bundles of a stem, the phloem is typically located on the outer side (toward the epidermis) and the xylem is on the inner side (toward the pith).

(NCERT Ed. 2025-26, 11th, Page No. 76)

109. (3)

The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee. One petal of its flower bears an uncanny resemblance to the female of the bee in size, colour and markings. The male bee is attracted to what it perceives as a female, 'pseudocopulates' with the flower, and during that process is dusted with pollen from the flower. When this same bee 'pseudocopulates' with another flower, it transfers pollen to it and thus, pollinates the flower.

(NCERT Ed. 2025-26, 12th, Page No. 202)

110. (4)

The genus *Drosophila* is for fruit fly. Housefly is *Musca domestica*.

(NCERT Ed. 2025-26, 11th, Page No. 8)

111. (2)

The ribosome also acts as a catalyst (23S rRNA in bacteria is the enzyme- ribozyme) for the formation of peptide bond. Before it was called transfer RNA, tRNA was known as soluble RNA (sRNA). In eukaryotes, the direct product of transcription from DNA is heterogeneous nuclear RNA (hnRNA). It is considered the primary transcript because it contains both exons (coding) and introns (non-coding). The mature messenger RNA (mRNA) in eukaryotes is produced only after the hnRNA undergoes processing, which includes splicing (removing introns), capping, and tailing.

(NCERT Ed. 2025-26, 12th, Page No. 95, 97, 99)

112. (2)

Species of *Sphagnum*, a moss, provide peat that have long been used as fuel, and as packing material for trans-shipment of living material because of their capacity to hold water.

(NCERT Ed. 2025-26, 11th, Page No. 29)

113. (1)

Vertical distribution of different species occupying different levels is called stratification. Species composition and stratification are the two main structural features of an ecosystem. For example, trees occupy top vertical strata or layer of a forest, shrubs the second and herbs and grasses occupy the bottom layers.

(NCERT Ed. 2025-26, 12th, Page No. 206)

114. (3)

The primary CO_2 acceptor in C_4 plants is a 3-carbon molecule phosphoenol pyruvate (PEP) and is present in the mesophyll cells. It acts as its primary acceptor, not a reducing agent.

(NCERT Ed. 2025-26, 11th, Page No. 146)

115. (2)

The Golgi apparatus is a single-membrane bound organelle. RER is frequently observed in the cells actively involved in protein synthesis and secretion. The lysosomes are membrane bound vesicular structures formed by the process of packaging in the golgi apparatus. Mitochondria are the sites of aerobic respiration.

(NCERT Ed. 2025-26, 11th, Page No. 95)

116. (1)

DNA is made of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases project inside. The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Similarly, Guanine is bonded with Cytosine with three H-bonds.

(NCERT Ed. 2025-26, 12th, Page No. 81)

117. (4)

In the process of glycolysis, ATP is utilized at two specific steps for every molecule of glucose:
Step 1: The conversion of Glucose to Glucose-6-phosphate.
Step 3: The conversion of Fructose-6-phosphate to Fructose-1,6-bisphosphate.

(NCERT Ed. 2025-26, 11th, Page No. 156)

118. (3)

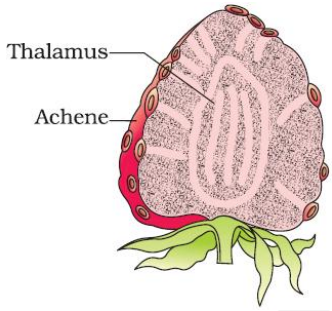
The flower is the reproductive unit in the angiosperms. It is meant for sexual reproduction. When a flower has both androecium and gynoecium, it is bisexual. Flowers with bracts-reduced leaf found at the base of the pedicel - are called bracteate and those without bracts, ebracteate. The arrangement of flowers on the floral axis is termed as inflorescence.

(NCERT Ed. 2025-26, 11th, Page No. 62)

[Contd...

119. (2)

The diagram shows a strawberry fruit (achenes on the surface with a fleshy thalamus). In most plants, by the time the fruit develops from the ovary, other floral parts degenerate and fall off. However, in a few species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called false fruits. Most fruits however develop only from the ovary and are called true fruits.



False fruit of strawberry

(NCERT Ed. 2025-26, 12th, Page No. 20)

120. (3)

In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. When the bulliform cells in the leaves have absorbed water and are turgid, the leaf surface is exposed.

(NCERT Ed. 2025-26, 11th, Page No. 77)

121. (3)

The cyanobacteria (also referred to as blue-green algae, a type of eubacteria) have chlorophyll *a* similar to green plants and are photosynthetic autotrophs. The cyanobacteria are unicellular, colonial or filamentous, freshwater/marine or terrestrial algae. The colonies are generally surrounded by gelatinous sheath. They often form blooms in polluted water bodies. Some of these organisms can fix atmospheric nitrogen in specialised cells called heterocysts, e.g., *Nostoc* and *Anabaena*.

(NCERT Ed. 2025-26, 11th, Page No. 13)

122. (1)

A bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. *Trichoderma* species are free-living fungi that are very common in the root ecosystems. They are effective biocontrol agents of several plant pathogens. Baculoviruses are pathogens that attack insects and other arthropods. The majority of baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*. Viruses are acellular organisms. An example of microbial biocontrol agents that can be introduced in order to control butterfly caterpillars is the bacteria *Bacillus thuringiensis* (often written as *Bt*).

(NCERT Ed. 2025-26, 12th, Page No. 157, 158)

123. (2)

Evolutionarily, the pteridophytes are the first terrestrial plants to possess vascular tissues – xylem and phloem. In pteridophytes, the main plant body is an independent sporophyte (diploid) which is differentiated into true root, stem and leaves.

(NCERT Ed. 2025-26, 11th, Page No. 31)

124. (3)

A normal human individual has 46 chromosomes, consisting of 22 pairs of autosomes (44 total) and 1 pair of sex chromosomes (XX or XY).

Klinefelter's syndrome is a genetic disorder caused by the presence of an additional X chromosome, resulting in a karyotype of 47, XXY. In this condition, the abnormality lies in the sex chromosomes; the number of autosomes remains exactly the same as in a normal individual (44 autosomes).

(NCERT Ed. 2025-26, 12th, Page No. 76)

125. (2)

One group of PGRs are involved in growth promoting activities. These are also called plant growth promoters, e.g., auxins, gibberellins and cytokinins. The PGRs of the other group play an important role in plant responses to wounds and stresses of biotic and abiotic origin. They are also involved in various growth inhibiting activities such as dormancy and abscission. The PGR abscisic acid belongs to this group. The gaseous PGR, ethylene, could fit either of the groups, but it is largely an inhibitor of growth activities.

(NCERT Ed. 2025-26, 11th, Page No. 174)

126. (4)

Less than 2% of the human genome codes for proteins. The total number of genes is estimated to be about 30,000 (much lower than previous estimates of 80,000 to 140,000). Repeated sequences make up very large portion of the human genome.

(NCERT Ed. 2025-26, 12th, Page No. 104)

[Contd...

127. (3)

Antonie Von Leeuwenhoek first saw and described a live cell. Robert Brown later discovered the nucleus.

(NCERT Ed. 2025-26, 11th, Page No. 100)

128. (1)

In a tree ecosystem, a single large tree (the producer) supports a large number of herbivorous birds or insects (primary consumers). If we then consider parasites living on those birds, the number increases even further at the higher trophic levels. Because the base starts with "1" (the tree) and the higher levels have many individuals, the pyramid is inverted.

(NCERT Ed. 2025-26, 12th, Page No. 213)

129. (2)

The single chlorophyll *a* molecule forms the reaction centre of the photosystem. Though chlorophyll is the major pigment responsible for trapping light, other thylakoid pigments like chlorophyll *b*, xanthophylls and carotenoids, which are called accessory pigments, also absorb light and transfer the energy to chlorophyll *a*. Indeed, they not only enable a wider range of wavelength of incoming light to be utilised for photosynthesis but also protect chlorophyll *a* from photo-oxidation.

(NCERT Ed. 2025-26, 11th, Page No. 138)

130. (4)

In Linnaeus' time a Two Kingdom system of classification with Plantae and Animalia kingdoms was developed that included all plants and animals respectively. This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms.

(NCERT Ed. 2025-26, 11th, Page No. 10)

131. (2)

The drones (male honey bee) are haploid organisms. Their cells do not undergo meiosis for gamete formation. Therefore, crossing over cannot occur during division in their cells.

(NCERT Ed. 2025-26, 12th, Page No. 71)

132. (3)

Both the species benefit in mutualism and both lose in competition in their interactions with each other. In amensalism on the other hand one species is harmed whereas the other is unaffected.

(NCERT Ed. 2025-26, 12th, Page No. 197)

133. (3)

Underground stems of potato, ginger, turmeric, zaminkand, *Colocasia* are modified to store food in them. They also act as organs of perennation to tide over conditions unfavourable for growth.

(NCERT Ed. 2022-23, 11th, Page No. 68)

134. (1)

Trait being inherited	Characteristic
Flower colour in dogflower	Incomplete dominance
Flower colour in pea	Complete dominance
Skin colour in humans	Polygenic inheritance
ABO blood grouping in humans	Multiple alleles

(NCERT Ed. 2025-26, 12th, Page No. 60, 61, 69)

135. (3)

Flowering plants have developed many devices to discourage self-pollination and to encourage cross-pollination. In some species, pollen release and stigma receptivity are not synchronised. Either the pollen is released before the stigma becomes receptive or stigma becomes receptive much before the release of pollen. In some other species, the anther and stigma are placed at different positions so that the pollen cannot come in contact with the stigma of the same flower. Both these devices prevent autogamy. The third device to prevent inbreeding is self-incompatibility. This is a genetic mechanism and prevents self-pollen (from the same flower or other flowers of the same plant) from fertilising the ovules by inhibiting pollen germination or pollen tube growth in the pistil. Another device to prevent self-pollination is the production of unisexual flowers. If both male and female flowers are present on the same plant such as castor and maize (monoecious), it prevents autogamy but not geitonogamy. In several species such as papaya, male and female flowers are present on different plants, that is each plant is either male or female (dioecy). This condition prevents both autogamy and geitonogamy.

(NCERT Ed. 2025-26, 12th, Page No. 15)

[Contd...

136. (3)

Members of Platyhelminthes have an incomplete digestive system. They possess a single opening (mouth) and lack an anus. Platyhelminthes are acoelomates — they do not have a body cavity at all.

(NCERT Ed. 2025-26, 11th, Page No. 38, 42)

137. (1)

It is the most abundant protein in animals, but not in the entire biosphere (e.g., RuBisCO is more abundant globally).

Lysozyme is a protein enzyme, not a nucleic acid (catalytic nucleic acids are ribozymes).

Competitive inhibitor competes with the substrate for the active site.

Arachidonic acid is a 20-carbon fatty acid.

Cellulose is made of β-D-glucose, not fructose.

(NCERT Ed. 2025-26, 11th, Page No. 106, 117)

138. (3)

In a cockroach, the alimentary canal is divided into foregut, midgut, and hindgut. A ring of 100-150 yellow-coloured thin filamentous Malpighian tubules is present at the junction of the midgut and hindgut. They help in the removal of excretory products from the haemolymph.

(NCERT Ed. 2020-21, 11th, Page No. 113)

139. (2)

Octopus is an example of Mollusca and scorpion is an example of arthropoda. They have a true coelom.

(NCERT Ed. 2025-26, 11th, Page No. 189)

140. (3)

List-I	List-II
Lub	First heart sound
Basophils	Secrete histamine, serotonin, and heparin
Dub	Associated with the closure of the semilunar valves
Lymphocytes	B and T forms responsible for immune responses

(NCERT Ed. 2025-26, 11th, Page No. 194, 195, 200)

141. (3)

Cardiac output is calculated as Stroke Volume x Heart Rate = 70 mL x 72 = 5040 mL (approx. 5 Litres). On average, 1100-1200 mL of blood is filtered by the kidneys per minute, which constitutes roughly 1/5th of the blood pumped out by each ventricle of the heart in a minute. Hence, 1100 mL is the correct answer. (125 mL is the GFR, not the blood volume).

(NCERT Ed. 2025-26, 11th, Page No. 208)

142. (2)

A Motor unit is formed by a motor neuron and the muscle fibres it supplies.

At the neuromuscular junction, Acetylcholine is released, not adrenaline.

Calcium ion binds to troponin, removing the blocking effect on actin active sites.

During muscle contraction in Sliding filament theory, the A band remains constant while the I band shortens.

The head of Myosin has ATPase activity, which provides energy for contraction.

(NCERT Ed. 2025-26, 11th, Page No. 221, 222)

143. (2)

The inner parts of cerebral hemispheres and a group of associated deep structures like the amygdala, hippocampus, etc., form a complex structure called the limbic lobe or limbic system. Along with the hypothalamus, it is involved in the regulation of sexual behaviour, expression of emotional reactions, and motivation.

(NCERT Ed. 2025-26, 11th, Page No. 236)

144. (1)

List-I	List-II
Erythropoietin	Juxtaglomerular cells of kidney
Gastrin	Stimulates secretion of HCl and pepsinogen
Secretin	Stimulates secretion of water and bicarbonate ions
Atrial natriuretic factor	Atrial wall of heart

(NCERT Ed. 2025-26, 11th, Page No. 247)

145. (1)

The rapid mid-cycle surge of Luteinizing hormone triggers rupture of the Graafian follicle, leading to ovulation.

The fundus is the dome-shaped upper portion of the uterus, located above the openings of the fallopian tubes.

The fallopian tube (oviduct) has four parts:

- Infundibulum
- Ampulla
- Isthmus
- Uterine (intramural) part.

(NCERT Ed. 2025-26, 12th, Page No. 29, 35)

[Contd...

146. (2)

'Saheli' is a once-a-week pill containing a non-steroidal preparation. It was developed by scientists at the Central Drug Research Institute (CDRI) in Lucknow, India.

(NCERT Ed. 2025-26, 12th, Page No. 43)

147. (3)

According to Hardy-Weinberg equilibrium, $p + q = 1$. Given q (recessive allele frequency) = 0.4. Therefore, $p = 1 - 0.4 = 0.6$. The dominant phenotype includes homozygous dominant (p^2) and heterozygous (2pq) individuals.

Percentage = $(p^2 + 2pq) \times 100$

$= (0.36 + 2(0.6)(0.4)) \times 100$

$= (0.36 + 0.48) \times 100$

$= 0.84 \times 100 = 84\%$.

(NCERT Ed. 2025-26, 12th, Page No. 121)

148. (1)

Typhoid is caused by *Salmonella typhi* and is confirmed by the Widal test.

Pneumonia, often caused by *Haemophilus influenzae,* leads to fluid accumulation in alveoli.

Plasmodium falciparum causes malignant and severe malaria.

Elephantiasis caused by *Wuchereria bancrofti* usually affects lower limbs and lymphatic regions rather than specifically the upper limbs.

(NCERT Ed. 2025-26, 12th, Page No. 130, 131, 133)

149. (2)

The correct sequence of a PCR cycle is Denaturation (heating to separate DNA strands), Annealing (cooling to allow primers to bind), and Extension (*Taq* polymerase synthesizes the new DNA strand).

(NCERT Ed. 2025-26, 12th, Page No. 172)

150. (1)

RNA interference is a defensive mechanism present in all eukaryotes. It relies on the presence of a complementary dsRNA that binds to the target mRNA, causing its silencing and preventing its translation into protein.

(NCERT Ed. 2025-26, 12th, Page No. 180)

151. (2)

Aschelminthes (roundworms) possess organ-system level of body organisation and are bilaterally symmetrical, triploblastic, and pseudocoelomate animals. Platyhelminthes are acoelomate, while Annelida and Arthropoda are true coelomates.

(NCERT Ed. 2025-26, 11th, Page No. 43)

152. (3)

Pyrimidines include Cytosine, Thymine, and Uracil. Cytosine is found in both DNA and RNA. Uracil is found only in RNA. Thymine (which is 5-methyl uracil) is exclusively found in DNA. Adenine is a purine.

(NCERT Ed. 2025-26, 11th, Page No. 106)

153. (4)

Compound epithelium mainly provides protection rather than secretion and absorption.

Cuboidal epithelium is found in glandular ducts and nephron tubules.

Ciliated epithelium is present in bronchioles and fallopian tubes to move substances.

Tight junction prevents leakage instead of enabling cytoplasmic communication.

Gap junction allows passage of ions and molecules between cells.

(NCERT Ed. 2025-26, 11th, Page No. 101, 102)

154. (3)

Residual Volume (RV) is the volume of air remaining in the lungs even after a forcible expiration. It averages 1100 mL to 1200 mL. Functional Residual Capacity (FRC) is the volume remaining after a normal expiration (ERV + RV).

(NCERT Ed. 2025-26, 11th, Page No. 187)

155. (3)

In an ECG, the P-wave represents atrial depolarisation, the QRS complex represents ventricular depolarisation, and the T-wave represents ventricular repolarisation (return of the ventricles from excited to normal state). The end of the T-wave marks the end of systole.

(NCERT Ed. 2025-26, 11th, Page No. 201)

156. (2)

The Counter-current mechanism in the Nephron helps maintain a high osmotic gradient in the medulla, which is essential for concentrating urine by reabsorbing water from the filtrate.

(NCERT Ed. 2025-26, 11th, Page No. 211)

157. (1)

The 8th, 9th, and 10th pairs of ribs are called vertebrochondral (false) ribs because they do not articulate directly with the sternum but join the 7th rib via hyaline cartilage. The 11th and 12th pairs are floating ribs.

(NCERT Ed. 2025-26, 11th, Page No. 225)

[Contd...

158. (4)

Impulse transmission across an Electrical synapse is faster than across a Chemical synapse because electrical signals pass directly between cells.

The diencephalon is a part of the forebrain (prosencephalon).

The forebrain consists of:

- Cerebrum
- Thalamus
- Hypothalamus

The thalamus and hypothalamus together form the diencephalon.

(NCERT Ed. 2025-26, 11th, Page No. 234, 236)

159. (2)

Iodine is essential for the normal rate of hormone synthesis in the thyroid. Its deficiency causes low synthesis of thyroid hormones (hypothyroidism) and leads to the enlargement of the gland (goitre). Graves' disease is a form of hyperthyroidism.

(NCERT Ed. 2025-26, 11th, Page No. 242)

160. (1)

During Oogenesis, the Secondary oocyte retains most of the cytoplasm from the primary oocyte.

The first meiotic division is unequal, producing a large secondary oocyte and a tiny first polar body.

(NCERT Ed. 2025-26, 12th, Page No. 32, 33)

161. (3)

LNG-20 and Progestasert are hormone-releasing IUDs. Lippes loop is a non-medicated IUD. CuT and Multiload 375 are copper-releasing IUDs.

(NCERT Ed. 2025-26, 12th, Page No. 44)

162. (3)

Adaptive radiation refers to the evolution of different species from a common ancestor in a given geographical area, spreading out to occupy different ecological niches.

When multiple adaptive radiations occur in an isolated area (e.g., Australian marsupials), it is called convergent evolution.

(NCERT Ed. 2025-26, 12th, Page No. 117, 118)

163. (4)

Cytokine barriers consist of interferons secreted by virus-infected cells, which confer viral resistance to adjacent healthy cells. Physical barriers include skin and mucus; physiological include acid and tears; cellular include PMNLs and macrophages.

(NCERT Ed. 2025-26, 12th, Page No. 135)

164. (2)

List-I	List-II
Lysozyme	Breaks down bacterial cell wall
Cellulase	Breaks down plant cell wall
Chitinase	Breaks down fungal cell wall
DNA ligase	Joins DNA fragments

(NCERT Ed. 2025-26, 12th, Page No. 165, 171)

165. (2)

The Bt toxin protein exists as an inactive protoxin. Once an insect ingests it, it is converted into an active form due to the alkaline pH of the gut, which solubilises the protein crystals. The activated toxin binds to midgut epithelial cells, creating pores and causing lysis.

(NCERT Ed. 2025-26, 12th, Page No. 179)

166. (2)

In Chordata, the characteristic features include:

- A dorsal hollow tubular nerve cord (A)
- A solid rod-like notochord present just below the nerve cord (B)
- Paired pharyngeal gill slits (C)

Thus, A represents the nerve cord and B represents the notochord.

(NCERT Ed. 2025-26, 11th, Page No. 46)

167. (3)

Amino acids, sugars, etc., are primary metabolites that play known roles in basic physiological processes. Anthocyanins (pigments), morphine (alkaloid), and lemon grass oil (essential oil) are secondary metabolites.

(NCERT Ed. 2025-26, 11th, Page No. 108)

168. (4)

Frogs possess special venous connections. The connection between the intestine and liver is the hepatic portal system, while the connection between the lower parts of the body and the kidney is the renal portal system.

(NCERT Ed. 2025-26, 11th, Page No. 82)

169. (3)

About 70% of CO_2 is transported as bicarbonate (HCO_3^-) with the help of the enzyme carbonic anhydrase. About 20-25% is carried by haemoglobin as carbamino-haemoglobin, and about 7% is carried in a dissolved state through plasma.

(NCERT Ed. 2025-26, 11th, Page No. 189)

170. (3)

Individuals with Blood Group AB have both A and B antigens on their RBCs and possess no natural antibodies against A or B in their plasma. Thus, they are universal recipients.

Blood Group A → Has anti-B antibodies in plasma

Blood Group B → Has anti-A antibodies

Blood Group O → Has both anti-A and anti-B antibodies

Blood Group AB → Has no anti-A or anti-B antibodies

(NCERT Ed. 2025-26, 11th, Page No. 195)

[Contd...

171. (4)

ADH or vasopressin from the neurohypophysis facilitates water reabsorption from latter parts of the tubule, thereby preventing diuresis.

(NCERT Ed. 2025-26, 11th, Page No. 212)

172. (1)

Fibrous joints (e.g., sutures of skull) are immovable.

Bones are connected by cartilage (e.g., intervertebral discs). Synovial joints joints have a synovial cavity containing synovial fluid for lubrication. The atlas–axis joint is a pivot joint, not a hinge joint. It allows rotational movement.

(NCERT Ed. 2025-26, 11th, Page No. 227)

173. (2)

During the resting state, the axonal membrane is comparatively more permeable to K^+ and nearly impermeable to Na^+ and negatively charged proteins. This differential permeability helps establish the resting membrane potential.

(NCERT Ed. 2025-26, 11th, Page No. 232)

174. (3)

Steroid hormones (like estradiol, cortisol, testosterone, progesterone) and iodothyronines cross the cell membrane and bind to intracellular receptors to regulate gene expression. Peptide and amino-acid derivative hormones (glucagon, insulin, epinephrine) bind to membrane-bound receptors.

(NCERT Ed. 2025-26, 11th, Page No. 248)

175. (1)

List-I	List-II
Leydig cells	Synthesise and secrete androgens
Sertoli cells	Provide nutrition to the male germ cells
Bulbourethral glands	Help in the lubrication of penis
Epididymis	Site of storage and maturation of sperms

(NCERT Ed. 2025-26, 12th, Page No. 27, 28, 32)

176. (1)

Medical termination of pregnancy (MTP) is considered relatively safe during the first trimester, i.e., up to 12 weeks of pregnancy. Second-trimester abortions are much riskier.

(NCERT Ed. 2025-26, 12th, Page No. 46)

177. (2)

Flippers of penguins (birds) and dolphins (mammals) have different anatomical origins but perform the same function (swimming). This indicates convergent evolution, making them analogous structures.

(NCERT Ed. 2025-26, 12th, Page No. 115)

178. (4)

It is a malady that can only be tackled, by the society and medical fraternity acting together, to prevent the spread of the disease.

In HIV/AIDS, infected macrophages continue to produce virus particles and act like an HIV factory, helping in persistent infection.

(NCERT Ed. 2025-26, 12th, Page No. 140)

179. (1)

Restriction enzymes like *Eco*RI cut the DNA backbone slightly away from the axis of symmetry (centre) of the palindrome, between the specific identical bases on opposite strands (G and A). This offset cutting is precisely what creates the single-stranded overhanging "sticky ends".

(NCERT Ed. 2025-26, 12th, Page No. 167)

180. (2)

As lymphocytes are not immortal cells, they die after their normal lifespan. Hence, the patient requires periodic infusions of fresh genetically engineered lymphocytes containing the functional ADA cDNA. A permanent cure would involve introducing the gene into bone marrow cells at early embryonic stages.

(NCERT Ed. 2025-26, 12th, Page No. 182)

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